

```

zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week05-service-setup$ dpkg -l nginx
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name          Version          Architecture Description
+++-+-----+-----+-----+-----+
ii  nginx           1.24.0-2ubuntu7.18 amd64        small, powerful, scalable web/proxy server
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week05-service-setup$ sudo apt purge nginx && sudo apt au
toremove -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  nginx-common
Use 'sudo apt autoremove' to remove it.
The following packages will be REMOVED:
  nginx*
0 upgraded, 0 newly installed, 1 to remove and 60 not upgraded.
After this operation, 1,355 kB disk space will be freed.
Do you want to continue? [Y/n] Y
(Reading database ... 126298 files and directories currently installed.)
Removing nginx (1.24.0-2ubuntu7.18) ...
Processing triggers for man-db (2.12.0-4build2) ...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  nginx-common
0 upgraded, 0 newly installed, 1 to remove and 60 not upgraded.
After this operation, 247 kB disk space will be freed.
(Reading database ... 126293 files and directories currently installed.)
Removing nginx-common (1.24.0-2ubuntu7.18) ...
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||/ Name          Version          Architecture Description
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un  nginx           <none>           <none>        (no description available)
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week05-service-setup$ |

```

Remove is used to delete the program but it will still keep configuration files present. On the other hand, purge is used to rid of both the program and configuration file.

```

Fetched 570 kB in 0s (1,564 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 126273 files and directories currently installed.)
Preparing to unpack ../nginx-common_1.24.0-2ubuntu7.18_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.18) ...
Selecting previously unselected package nginx.
Preparing to unpack ../nginx_1.24.0-2ubuntu7.18_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.18) ...
Setting up nginx-common (1.24.0-2ubuntu7.18) ...
Setting up nginx (1.24.0-2ubuntu7.18) ...
* Upgrading binary nginx
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
systemctl restart systemd-logind.service

No containers need to be restarted.

User sessions running outdated binaries:
zainwright @ session #1: sshd[1381]
zainwright @ user manager service: systemd[1390]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
Starting and enabling Nginx service...
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-installer.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
Verifying Nginx status...
SUCCESS: Nginx is up and running successfully!
Server setup completed successfully.
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week05-service-setup$ systemctl is-active nginx
active
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week05-service-setup$ systemctl is-enabled nginx
enabled
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week05-service-setup$

```

While start and enable may come off as the same they are pretty different. Start is used in the middle of a session and will not be started again until asked to during the next session. Enable will have it start every time a session boots and will be ready for you as you start a new session.

```
GNU nano 7.2 /home/zainwright/server-setup.sh *
#!/bin/bash

#####

# Script Name: server-setup.sh

# Purpose: Automate the installation, enabling, and confirmation of Nginx.

# Author: Zain Wright

# Date: September 26th, 2026

# Usage: ./server-setup.sh

#####

echo "Starting server setup..."

# Fetch the latest package lists from the repositories to ensure install of the newest versions
echo "Updating package repositories..."
sudo apt update

# Install Nginx. The '-y' flag automatically answers 'yes' to prompts for us, allowing unattended installation
echo "Installing Nginx web server..."
sudo apt install -y nginx

# Enable Nginx to start on boot and '--now' starts the service immediately so we don't have to run a command manually
echo "Starting and enabling Nginx service..."
sudo systemctl enable --now nginx

# Verification block
echo "Verifying Nginx status..."
if systemctl is-active --quiet nginx; then
    echo "SUCCESS: Nginx is up and running successfully!"
else
    echo "ERROR: Nginx failed to start or is not running properly."
    exit 1
fi

echo "Server setup completed successfully."
|
```

```
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week05-service-setup$ sudo apt purge nginx && sudo apt au
toremove -y
[sudo] password for zainwright:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
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(Reading database ... 126293 files and directories currently installed.)
Removing nginx-common (1.24.0-2ubuntu7.18) ...
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week05-service-setup$ systemctl status nginx
Unit nginx.service could not be found.
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week05-service-setup$ sudo bash ~/server-setup.sh
Starting server setup...
Updating package repositories...
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
60 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing Nginx web server...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
awk -F:
  Flags: -F: (set field separator to colon)
  Example: awk -F: '{print $1, $3}' /etc/passwd
  Risk: Parsing sensitive files like /etc/passwd requires caution to avoid exposing user details.

apt update
  Flags: none
  Example: sudo apt update
  Risk: Missing this step before installing new packages can cause dependency conflicts or install out of date packages.

apt install -y
  Flags: -y (assume yes to all prompts)
  Example: sudo apt install -y nginx
  Risk: Automatically accepting installations skips the prompt displaying how much disk space will be used.

apt purge
  Flags: none
  Example: sudo apt purge nginx
  Risk: Deletes the package along with all its custom configuration files, which can cause permanent loss of configuration.

apt autoremove
  Flags: none
  Example: sudo apt autoremove
  Risk: Might occasionally remove orphan library packages that other custom, compiled-from-source tools depend on.

dpkg -l
  Flags: -l (list matching packages)
  Example: dpkg -l | grep nginx
  Risk: Generating un-filtered lists prints massive amounts of text to the CLI screen, making it easy to miss important information.

systemctl lifecycle commands
  Flags: start, stop, restart, status
  Example: sudo systemctl restart nginx
```